WK1-PM1 – Cybersecurity Lab Setup Report

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Task	WK1-PM1 – Lab Setup (Week 1, Project Module 1)
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Objective

The goal of this task is to build a personal cybersecurity testing lab with Kali Linux as the attacking machine. Kali sits on an isolated NAT network in the 10.0.0.0/24 subnet, uses the static address 10.0.0.2/24, and still has full Internet access through the NAT gateway. The VM also needs clipboard sharing, drag-and-drop, and a shared Downloads folder from the host.

Lab environment

The lab was built on VMware Workstation instead of VirtualBox. Every network requirement was mapped to the VMware equivalent, shown in the table below.

Item	Configuration
Host OS	Windows 11
Hypervisor	VMware Workstation (version: *fill in*)
Guest OS	Kali Linux, installed from ISO (version: *fill in*)
VM resources	6 GB RAM, 4 vCPU, 40 GB disk
Virtual network	VMnet8 (NAT), subnet 10.0.0.0/24
NAT gateway	10.0.0.1
DHCP pool	10.0.0.128 – 10.0.0.254
Kali address	10.0.0.2/24 (static), DNS 8.8.8.8

VirtualBox to VMware mapping

Requirement (VirtualBox)	Equivalent used (VMware Workstation)
NAT Network 10.0.0.0/24	VMnet8 (NAT) set to 10.0.0.0/24 in Virtual Network Editor
Adapter attached to NAT Network	Network Adapter → Custom: VMnet8 (NAT)
Shared Clipboard & Drag'n'Drop	Options → Guest Isolation (copy/paste and drag-and-drop)
Shared Folders	Options → Shared Folders (mounted in Kali under /mnt/hgfs)

Requirements compliance checklist

#	Requirement	Status	Evidence
1	Use VirtualBox as base	Done with VMware Workstation (see mapping above)	Fig. 1, 5
2	Kali Linux as attacking machine	Done	Fig. 1, 6, 7
3	Network in subnet 10.0.0.0/24	Done	Fig. 2
4	NAT network with 10.0.0.0/24	Done (VMnet8 NAT)	Fig. 2, 3, 4, 5
5	Clipboard & drag/drop enabled	Done	Step 7
6	Shared folder: host Downloads	Done	Step 8
7	Kali IP 10.0.0.2/24	Done	Fig. 6, 7
8	Kali has full Internet access	Done	Fig. 7

Implementation steps

Step 1 – Create the Kali Linux VM

Kali Linux was installed from the ISO in VMware Workstation with 6 GB RAM, 4 processors and a 40 GB disk. By default, VMware attached the network adapter to the standard NAT mode.

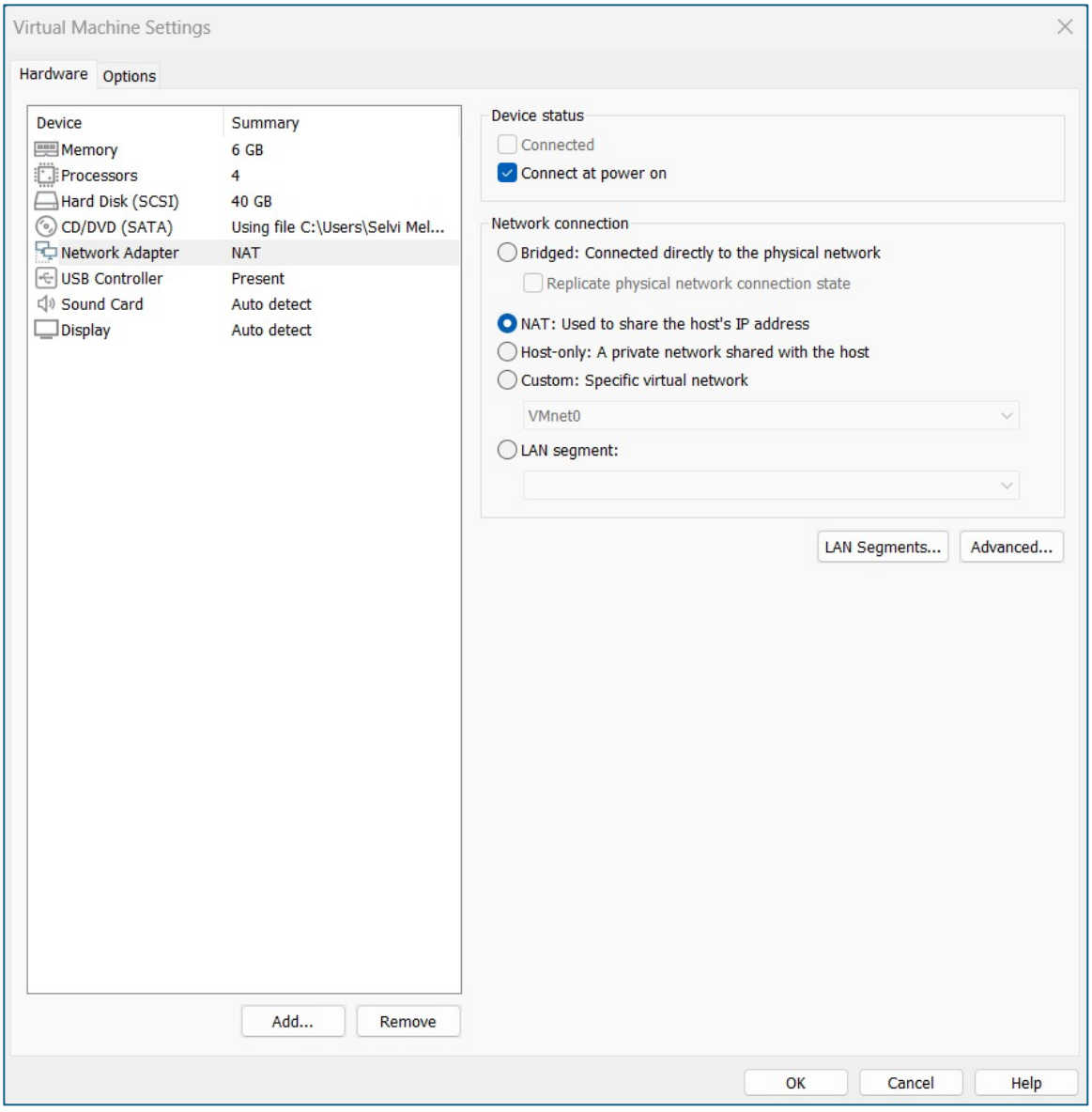


Figure 1 – Kali VM hardware settings with the default NAT adapter

Step 2 – Configure the NAT network in 10.0.0.0/24

In Edit → Virtual Network Editor, VMnet8 (NAT) was changed to subnet 10.0.0.0 with mask 255.255.255.0. This plays the same role as the VirtualBox NAT Network.

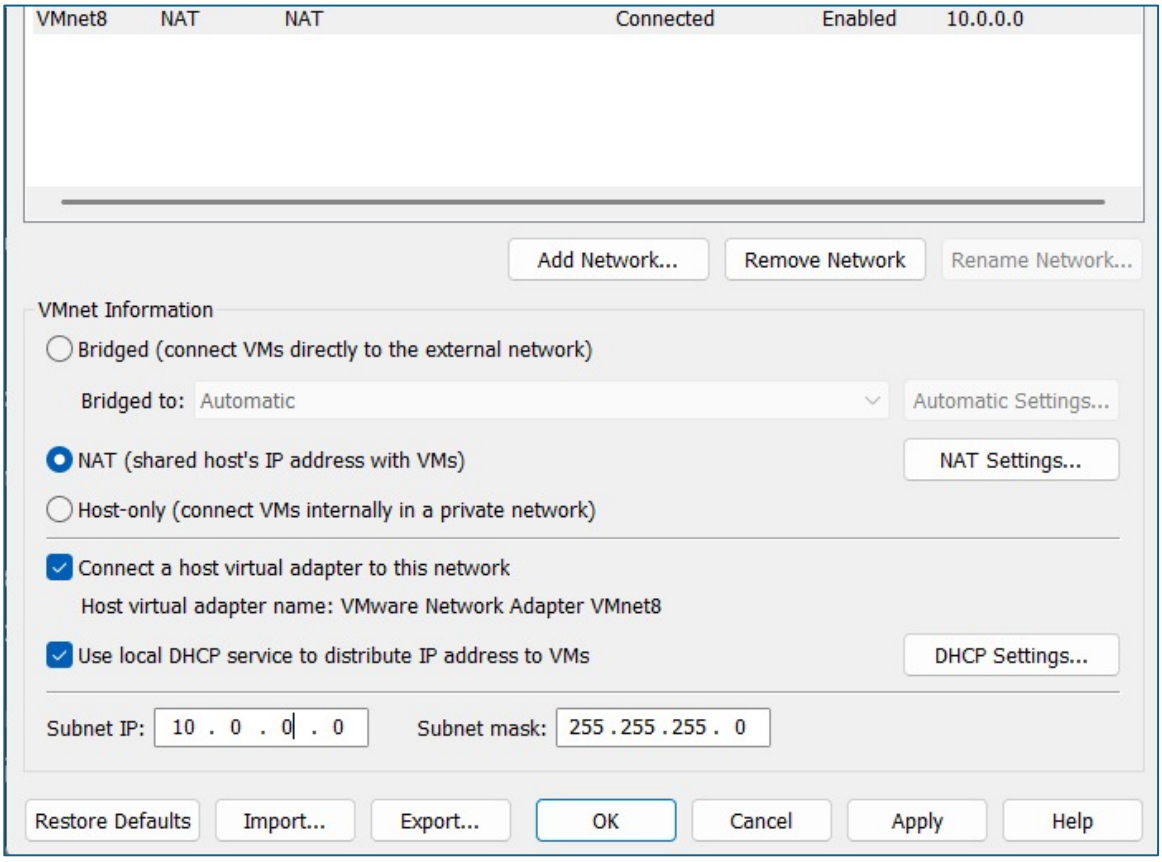


Figure 2 – VMnet8 (NAT) set to subnet 10.0.0.0/24

Step 3 – Keep DHCP away from the static range

The DHCP pool for VMnet8 hands out 10.0.0.128 – 10.0.0.254. Low addresses such as 10.0.0.2 stay free for static assignment, so Kali's address never collides with a DHCP lease.

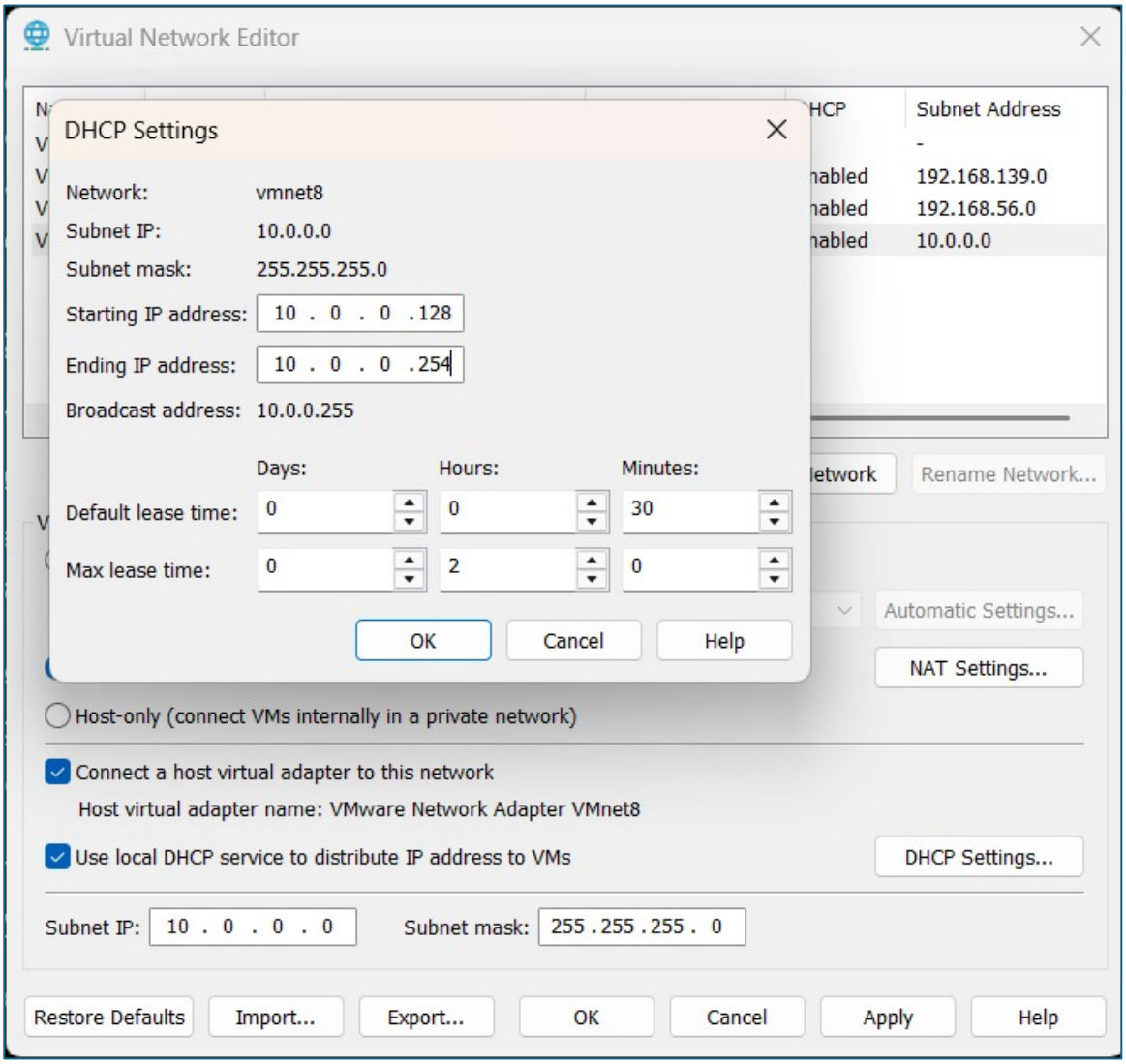


Figure 3 – VMnet8 DHCP range 10.0.0.128 – 10.0.0.254

Step 4 – Set the NAT gateway to 10.0.0.1

In NAT Settings, the gateway IP was set to 10.0.0.1 to match the lab design, where 10.0.0.1 is the default gateway for all VMs.

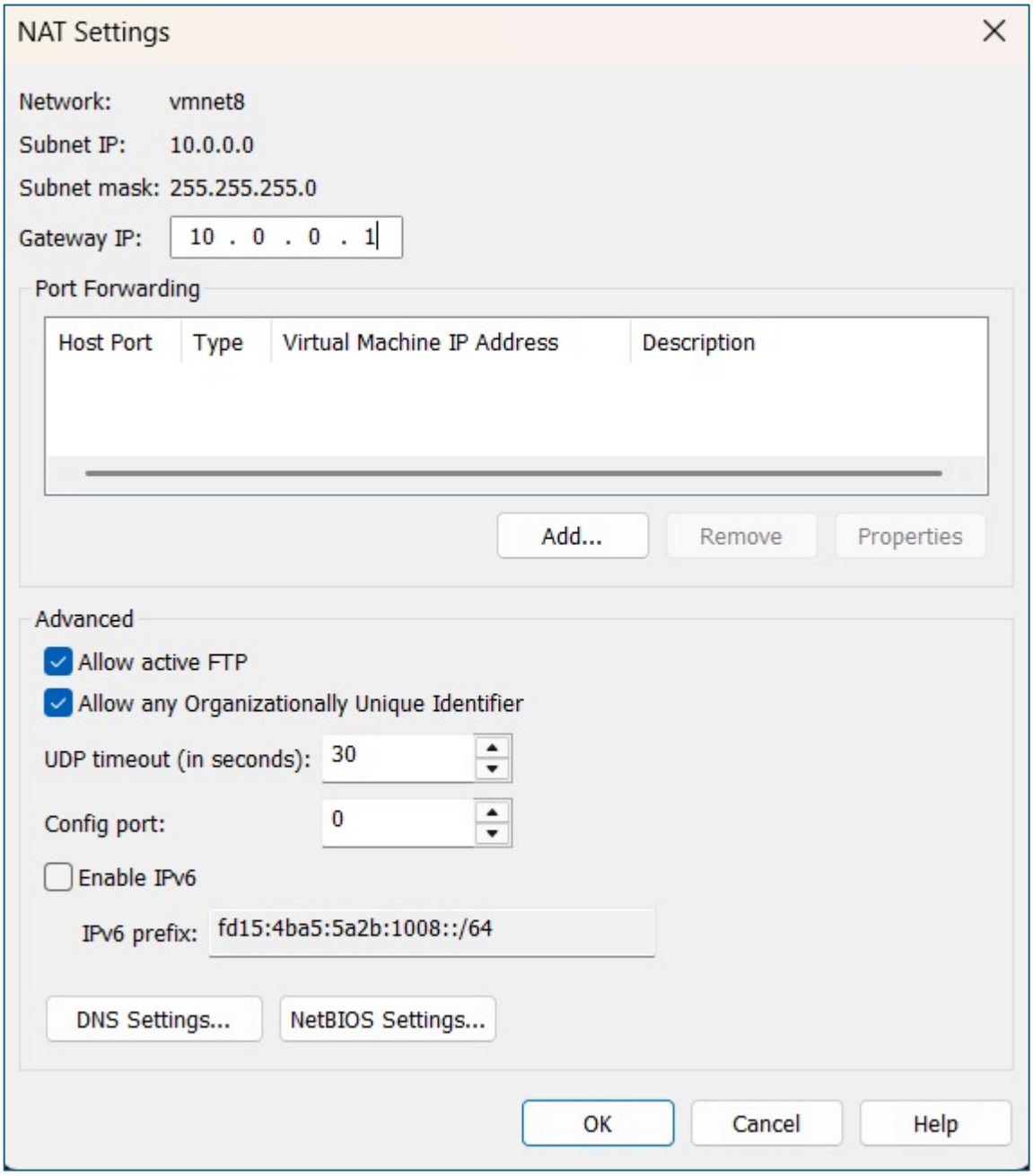


Figure 4 – NAT Settings with gateway 10.0.0.1

Step 5 – Attach the Kali adapter to VMnet8

The Kali network adapter was switched to Custom → VMnet8 (NAT), so the VM joins the 10.0.0.0/24 network explicitly.

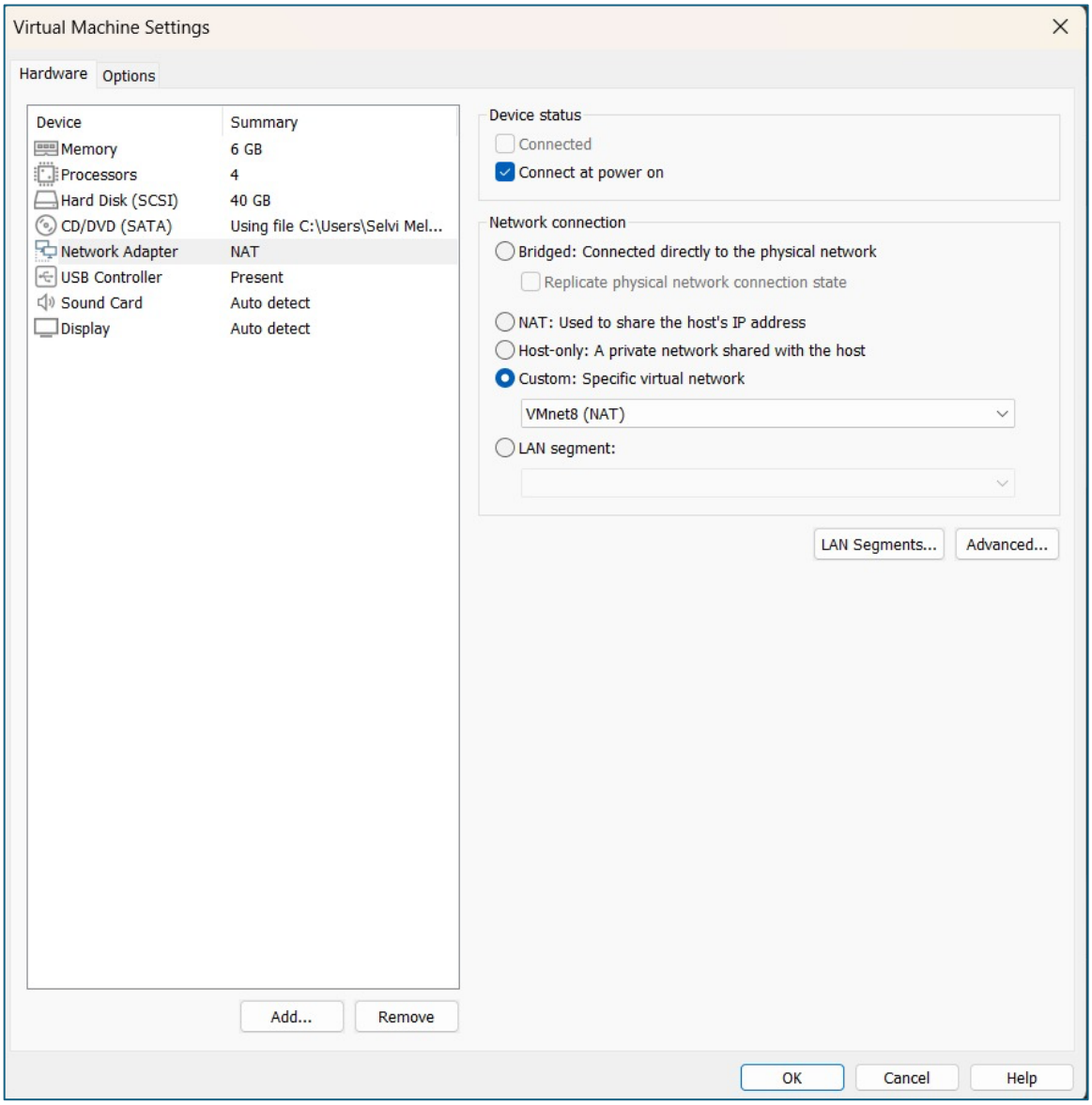


Figure 5 – Kali adapter attached to VMnet8 (NAT)

Step 6 – Set Kali's static IP 10.0.0.2/24

Inside Kali, nmtui was used to edit "Wired connection 1". IPv4 was set to Manual with address 10.0.0.2/24, gateway 10.0.0.1 and DNS 8.8.8.8.

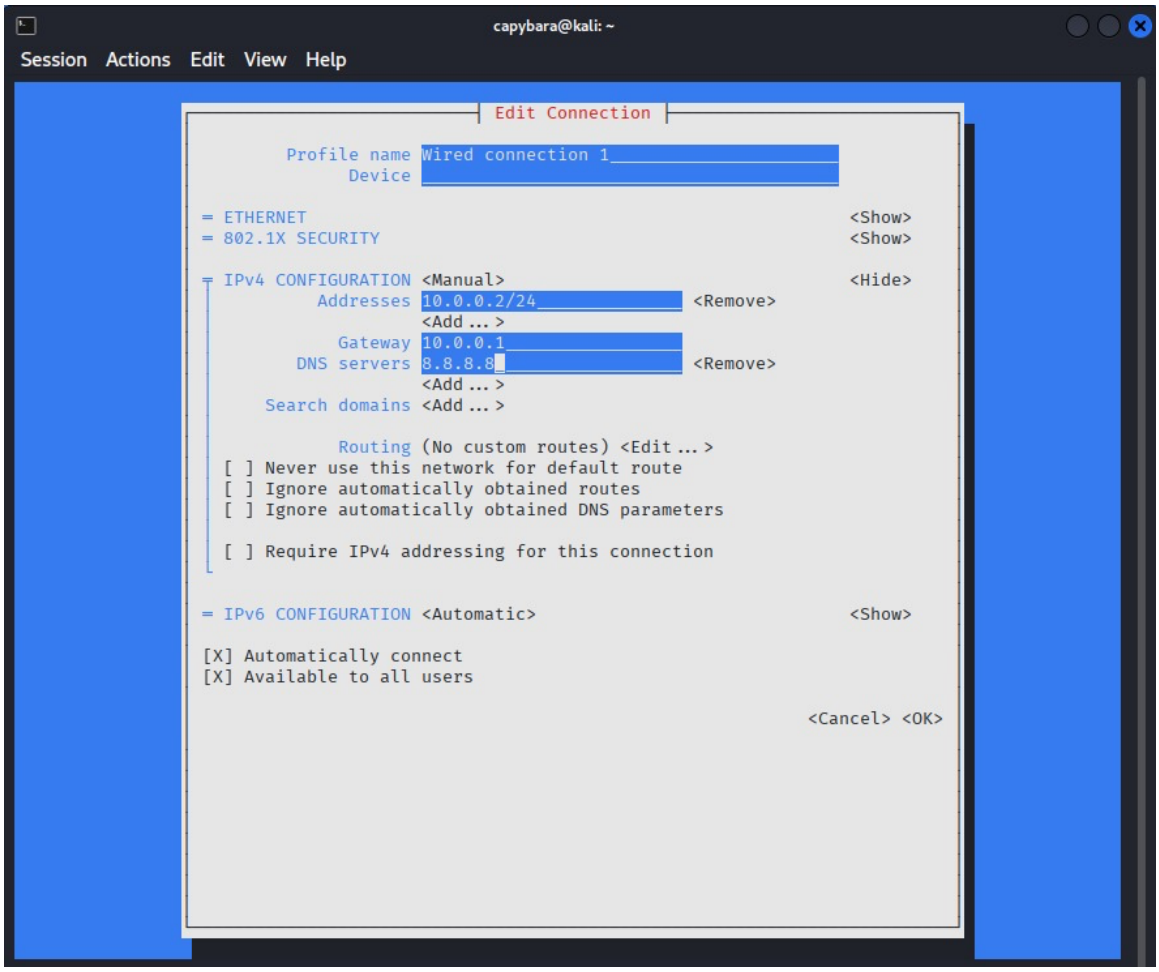


Figure 6 – Manual IPv4 configuration in nmtui

The connection was then reactivated with sudo nmcli connection up "Wired connection 1". The output of ip -br addr shows eth0 UP with 10.0.0.2/24, and ip route shows the default route via 10.0.0.1.

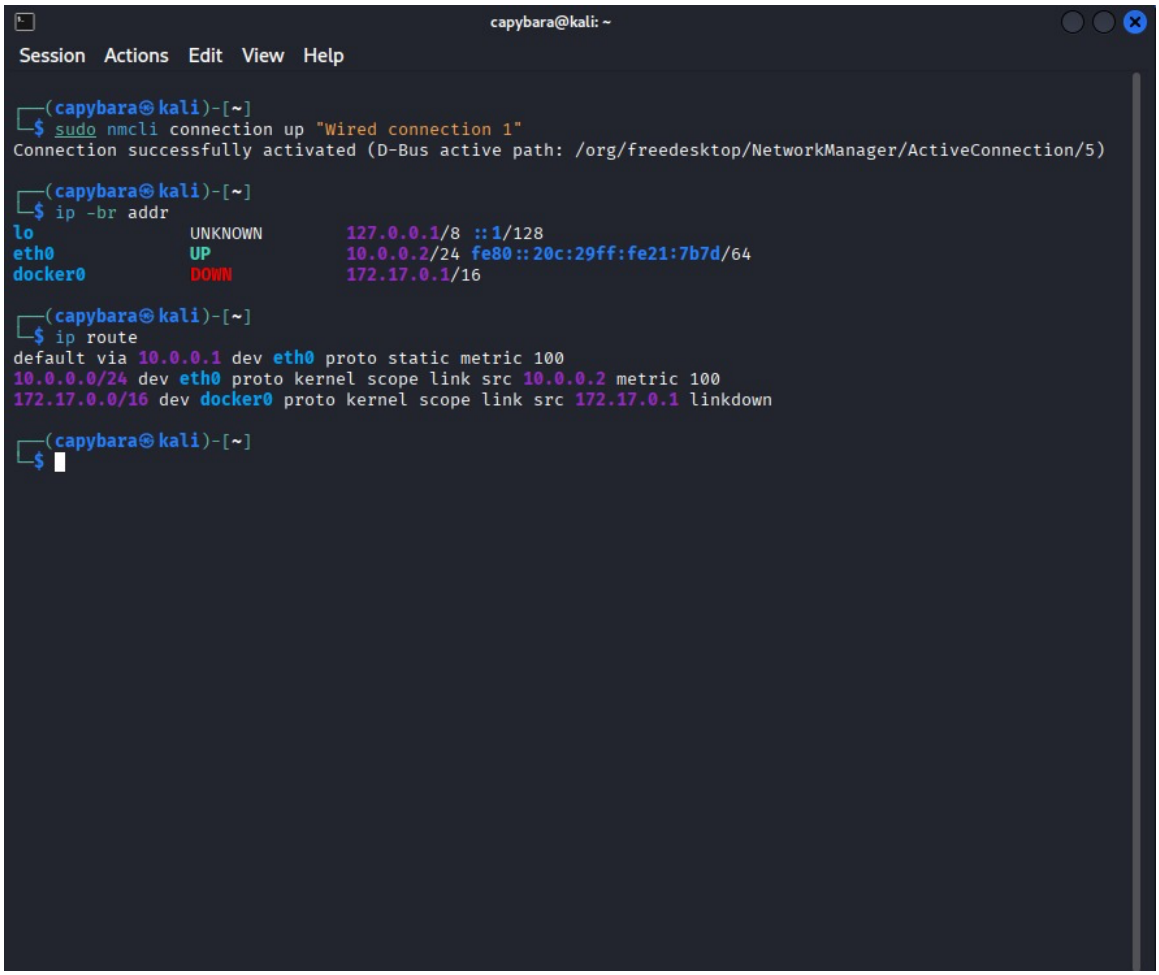


Figure 7 – eth0 holds 10.0.0.2/24, default route via 10.0.0.1

Step 7 – Enable clipboard and drag-and-drop

With the VM powered off, VM Settings → Options → Guest Isolation was opened and both "Enable drag and drop" and "Enable copy and paste" were ticked. After booting Kali, text copied on the host could be pasted into the Kali terminal and back, and a file dragged from the host desktop landed inside the VM.

Step 8 – Share the host Downloads folder

Under VM Settings → Options → Shared Folders, sharing was set to "Always enabled" and the host Downloads folder was added. Inside Kali, the share is mounted under /mnt/hgfs, and `ls /mnt/hgfs/Downloads` listed the files from the host's Downloads folder.

Verification

The tests below confirm Kali's address, routing, Internet access, shared folder and clipboard.

Test	Command / action	Result
IP address	`ip -br addr`	eth0 UP, 10.0.0.2/24 (Fig. 7)
Default route	`ip route`	default via 10.0.0.1 (Fig. 7)
Gateway reachable	`ping -c 4 10.0.0.1`	Success, replies received
Internet by IP	`ping -c 4 8.8.8.8`	Success, replies received
DNS + Internet	`ping -c 4 google.com`	Success, name resolved and replies received
Shared folder	`ls /mnt/hgfs/Downloads`	Success, host files listed
Clipboard	Copy text host ↔ Kali	Success, both directions
Drag-and-drop	Drag a file from host into Kali	Success, file copied into the VM

All tests passed, so every requirement of WK1-PM1 is met.

Issues & notes

- **VMware instead of VirtualBox.** The task asks for VirtualBox. VMware Workstation was used, and each VirtualBox setting was mapped to its VMware equivalent (see the mapping table). The network design, subnet, gateway and Kali address match the task exactly.
- **Gateway address in VMware.** VMware normally gives the host adapter (VMnet8) the .1 address and the NAT gateway .2. Here the gateway was moved to 10.0.0.1 so that 10.0.0.2 is free for Kali, as the task requires.
- **Static IP outside the DHCP pool.** Kali uses 10.0.0.2, while DHCP only leases .128 – .254. This avoids address conflicts when more VMs join later (e.g. Windows 10 at 10.0.0.10).
- **Connectivity fix.** If Kali has no Internet after setting the static IP, the course's fix applies set `ipv4.dad-timeout 0` on "Wired connection 1", then bring the connection down and up with nmcli.

Conclusion

The Kali Linux attacking machine runs on an isolated NAT network (10.0.0.0/24) with the static address 10.0.0.2/24, gateway 10.0.0.1 and full Internet access. Clipboard sharing, drag-and-drop and the shared Downloads folder all work. The lab is ready for Phase 2, where Windows and Android targets will join the same network.